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FURTHER MATHEMATICS

FM04

(9665/FM04) Unit FS2 Statistics

Mark scheme

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Key to mark scheme abbreviations

M	Mark is for method
m	Mark is dependent on one or more M marks and is for method
A	Mark is dependent on M or m marks and is for accuracy
B	Mark is independent of M or m marks and is for method and accuracy
E	Mark is for explanation
√or ft	Follow through from previous incorrect result
CAO	Correct answer only
CSO	Correct solution only
AWFW	Anything which falls within
AWRT	Anything which rounds to
ACF	Any correct form
AG	Answer given
SC	Special case
oe	Or equivalent
A2, 1	2 or 1 (or 0) accuracy marks
–*x* EE	Deduct *x* marks for each error
NMS	No method shown
PI	Possibly implied
SCA	Substantially correct approach
sf	Significant figure(s)
dp	Decimal place(s)

Q	Answer	Marks	Comments
1	TT median 0 TH or HT median 5 TT median 10 m: 0, 5, 10 $P(M = m)$: $\frac{1}{4}$, $\frac{1}{2}$, $\frac{1}{4}$	M1 m1 A1	Identify at least one possible combination and median Identify all possible combinations and medians Accept in function form
	Total	**3**	

Q	Answer	Marks	Comments
2(a)	Relative Efficiency = $\dfrac{\frac{100}{\sigma^2}}{\frac{10}{\sigma^2}}$ = 10 Linda's estimator is more efficient than Andrew's estimator	M1 A1 E1	Uses relative efficiency formula oe oe
2(b)(i)	$\mathrm{E}(P) = \dfrac{E\left(\overline{X_1}\right) + E\left(\overline{X_2}\right)}{2} = \dfrac{\mu + \mu}{2}$ = μ therefore unbiased	M1 A1	
2(b)(ii)	$\mathrm{Var}(P) = \dfrac{Var\left(\overline{X_1}\right) + Var\left(\overline{X_2}\right)}{2^2} = \dfrac{\frac{\sigma^2}{10} + \frac{\sigma^2}{100}}{2^2}$ = $\dfrac{11\sigma^2}{400}$	M1 A1	 oe
	Total	**7**	

Q	Answer	Marks	Comments
3(a)	$\bar{x} = 66.404$	B1	
	$s^2 = \frac{1}{500-1}\left(2210000 - \frac{33202^2}{500}\right)$	M1	Attempt at variance formula Allow one slip Implied by correct answer
	= 10.5(2984369)	A1	AWRT 10.5
	z = 2.0537	B1	AWRT 2.054
	$66.404 \pm 2.0537\sqrt{\frac{10.52...}{500}}$	M1	Applies confidence interval formula with their values
	(66.106, 66.702)	A1	AWRT
3(b)	Rejects as 67 is outside the confidence interval	E1ft	Follow through their confidence interval
3(c)	Still valid as sample size is large enough	E1	oe
	for central limit theorem to apply	E1	oe
	Total	**9**	

Q	Answer	Marks	Comments
4(a)(i)	P($X \leq 25$) = 0.9427 P($X \leq 26$) = 0.9686	M1	Both seen
	Accept if $X \leq 26$	M1	oe
	B(50, 0.5)	M1	Use of PI
	Type II probability = 0.664	A1	AWRT
4(a)(ii)	Power = 1 – 0.664	M1	Applies power formula with their Type II probability
	= 0.336	A1F	AWRT
4(b)	The power of the test would increase	E1	
	Total	**7**	

Q	Answer	Marks	Comments
5	H_0: $\sigma_A = \sigma_B$ H_1: $\sigma_A \neq \sigma_B$	B1	Both hypotheses
	dof 20, 25	B1	
	F = $\frac{s_A^2}{s_B^2} = \frac{0.3^2}{0.2^2}$	M1	
	= 2.25	A1	
	$F_{20,25}$ = 2.70	B1	AWRT
	Accept H_0	A1ft	Follow through their F and $F_{20,25}$ Implied by correct conclusion in context
	Evidence to suggest/support that the population variances of machines *A* and *B* are equal	E1	Must be consistent with their conclusion on whether to accept H_0 or not or their F and $F_{20,25}$ if not explicitly stated
	Total	**7**	

Q	Answer	Marks	Comments
6	H_0: There is no association between the office an employee works in and the transport they use to get to work H_1: There is an association between the office an employee works in and the transport they use to get to work	B1	Both hypotheses, variables must be referenced in at least the null hypothesis oe
		M1	At least three values AWRT PI
		A1	All values PI
		M1	
	dof v = 2	B1	
	$\sum\frac{(O-E)^2}{E}=\frac{(4-13.6)^2}{13.6}+\frac{(22-12.4)^2}{12.4}+\frac{(32-24.5...)^2}{24.5...}+\frac{(15-22.4...)^2}{22.4...}+\frac{(32-29.8...)^2}{29.8...}+\frac{(25-27.1...)^2}{27.1...}$	M1	Attempts $\sum\frac{(O-E)^2}{E}$ using their table even if columns not combined
	= 19.2 (B+T combined) or = 17.9 (C+T combined) or = 21.3 (W+T combined)	A1	Accept 19.1 AWRT

	W	C	B	T
N	13.6	24.6	26.2	3.66
S	12.4	22.4	23.8	3.34

	B+T Expected	B+T Actual
N	29.8	32
S	27.2	25

or

	C+T Expected	C+T Actual
N	28.2	34
S	25.7	20

or

	W+T Expected	W+T Actual
N	17.3	6
S	15.7	27

Combine two columns for expected and actual values

	Test statistic = 9.2 Reject H_0 Evidence to suggest/support that there is an association between the office an employee works in and the transport they use to get to work	**B1** **A1ft** **E1**	AWRT Follow through their $\sum \frac{(O-E)^2}{E}$ and test statistic Implied by correct conclusion in context Must be consistent with their conclusion on whether to accept H_0 or not or their $\sum \frac{(O-E)^2}{E}$ and test statistic if not explicitly stated
	Total	**10**	

Q	Answer	Marks	Comments
7(a)	H_0: $\mu_A = \mu_B$ H_1: $\mu_A \neq \mu_B$	B1	Both hypotheses
	$z = \dfrac{\lvert 504-502 \rvert}{\sqrt{\dfrac{2^2}{10}+\dfrac{3^2}{12}}}$	M1 M1	Numerator Denominator
	= 1.87	A1	AWRT
	z_{crit} = 1.96 or -1.96	B1	Seen anywhere
	Accept H_0	A1ft	Follow through their z and z_{crit} Implied by correct conclusion in context
	Evidence to suggest/support that the mean mass of jars of honey is the same for both suppliers	E1	Must be consistent with their conclusion on whether to accept H_0 or not or their z and z_{crit} if not explicitly stated
7(b)	t test/values instead of z test/values	E1	oe
	with dof ν = 20	E1	oe
	Calculate pooled estimate of variance S_P^2	E1	oe
	Replace $\dfrac{\sigma_A^2}{n_A}+\dfrac{\sigma_B^2}{n_B}$ with $s_p^2\left(\dfrac{1}{n_A}+\dfrac{1}{n_B}\right)$	E1	oe
	Total	**11**	

Q	Answer	Marks	Comments
8(a)	H_0: $\mu_B = \mu_A$ H_1: $\mu_B < \mu_A$	B1	Both hypotheses
	Student / Difference 1 / 4 2 / 1 3 / 0 4 / -1 5 / 2 6 / 3	B1	All differences found
	$\bar{x} = 1.5$	B1	
	$s^2 = \frac{1}{6-1}\left(31 - \frac{9^2}{6}\right)$	M1	Attempt at variance formula Allow one slip Implied by correct answer
	= 3.5	A1	Accept AWRT s = 1.87
	dof v = 5	B1	
	$t = \frac{1.5}{\sqrt{\frac{3.5}{6}}}$	M1	Using their s^2
	= 1.96	A1	AWRT
	t_5 cv = 2.02	B1	AWRT
	Accept H_0	A1ft	Follow through their t and t_5 Implied by correct conclusion in context
	No evidence to suggest/support that the tutorial company's claim is true	E1	Must be consistent with their conclusion on whether to accept H_0 or not or their t and t_5 if not explicitly stated
8(b)	The difference between test scores before and after the revision course	E1	oe
	are normally distributed	E1	oe
	Total	**13**	

Q	Answer	Marks	Comments
9(a)	$M'_X(t) = \ldots e^t(1 - 0.8e^t)^{-3}$	M1	$M'_X(t) = Ae^t(1 - 0.8e^t)^{-3}$ where A is a constant
	$M'_X(t) = 0.064e^t(1 - 0.8e^t)^{-3}$	A1	oe May be unsimplified
	Mean = $M'_X(0) = 0.064e^0(1 - 0.8e^0)^{-3}$	M1	Attempts $M'_X(0)$
	= 8	A1	
9(b)	$M''_X(t) = \ldots e^t(1 - 0.8e^t)^{-3} + \ldots e^{2t}(1 - 0.8e^t)^{-4}$	M1	$M''_X(t) = Ae^t(1 - 0.8e^t)^{-3} + Be^{2t}(1 - 0.8e^t)^{-4}$ where A and B are constants oe
	$M''_X(t) = 0.064e^t(1 - 0.8e^t)^{-3} + 0.1536e^{2t}(1 - 0.8e^t)^{-4}$	A1	oe May be unsimplified
	Variance = $M''_X(0) - (M'_X(0))^2$ $= 0.064e^0(1 - 0.8e^0)^{-3} + 0.1536e^{2\times 0}(1 - 0.8e^0)^{-4} - 8^2$	M1	Applies variance formula with their $M''_X(t)$ and mean
	= 40	A1	
9(c)	$M_{2+3X}(t) = e^{2t} M_X(3t)$	M1	
	$= e^{2t} \times 0.04(1 - 0.8e^{3t})^{-2}$	M1	
	$= \left(\dfrac{0.2e^t}{1-0.8e^{3t}}\right)^2$	A1	
9(d)	$M_{X+Y}(t) = M_X(t)\, M_Y(t)$ $= 0.04(1 - 0.8e^t)^{-2}(0.8 + 0.2e^t)^2$	M1	
	$= \left(\dfrac{0.16+0.04e^t}{1-0.8e^t}\right)^2$	A1	
	Total	**13**	